

Luiss

Libera Università Internazionale
degli Studi Sociali Guido Carli

Recursion

Introduction to Computer Programming

Management and Artificial Intelligence

LUISS



Preliminaries

Recursion is a powerful programming technique where a function calls itself to **solve smaller instances of the same problem**. It's a fundamental concept in computer science and programming, enabling the creation of elegant and efficient solutions to complex problems.

Preliminaries

Recursion is a powerful programming technique where a function calls itself to **solve smaller instances of the same problem**. It's a fundamental concept in computer science and programming, enabling the creation of elegant and efficient solutions to complex problems.

To understand recursion properly, we need to first explore two fundamental data structures:

1. **Stacks**: Essential for tracking recursive function calls
2. **Queues**: Similar to Stacks, but with a different order of operations

Stacks

The **stack** is one of the most popular abstract data types in computer science.

Stacks

The **stack** is one of the most popular abstract data types in computer science.

A stack is a sequence (collection) of elements where only the *last* element (i.e., the one at the *top* of the stack) can be accessed.

Stacks

The **stack** is one of the most popular abstract data types in computer science.

A stack is a sequence (collection) of elements where only the *last* element (i.e., the one at the *top* of the stack) can be accessed.

The allowed operations on a stack $\mathcal{S}$ are the following:

- $\text{push}(e, \mathcal{S})$: adds a new element $e \in V$, for any domain V , at the top of the stack $\mathcal{S}$
- $\text{pop}(\mathcal{S})$: returns and removes the current element at the top of the stack $\mathcal{S}$.

Stacks

The **stack** is one of the most popular abstract data types in computer science.

A stack is a sequence (collection) of elements where only the *last* element (i.e., the one at the *top* of the stack) can be accessed.

The allowed operations on a stack $\mathcal{S}$ are the following:

- $\text{push}(e, \mathcal{S})$: adds a new element $e \in V$, for any domain V , at the top of the stack $\mathcal{S}$
- $\text{pop}(\mathcal{S})$: returns and removes the current element at the top of the stack $\mathcal{S}$.

Optionally, it is possible to define an additional operation:

- $\text{peek}(\mathcal{S})$: reads the top element without modifying the stack.

A special domain value is $\mathcal{S}_\emptyset$, namely the empty stack.

A special domain value is $\mathcal{S}_\emptyset$, namely the empty stack.

As $\mathcal{S}_\emptyset$ has no elements:

- $\text{pop}()$ is not allowed
- $\text{peek}()$ is not allowed.

A special domain value is $\mathcal{S}_\emptyset$, namely the empty stack.

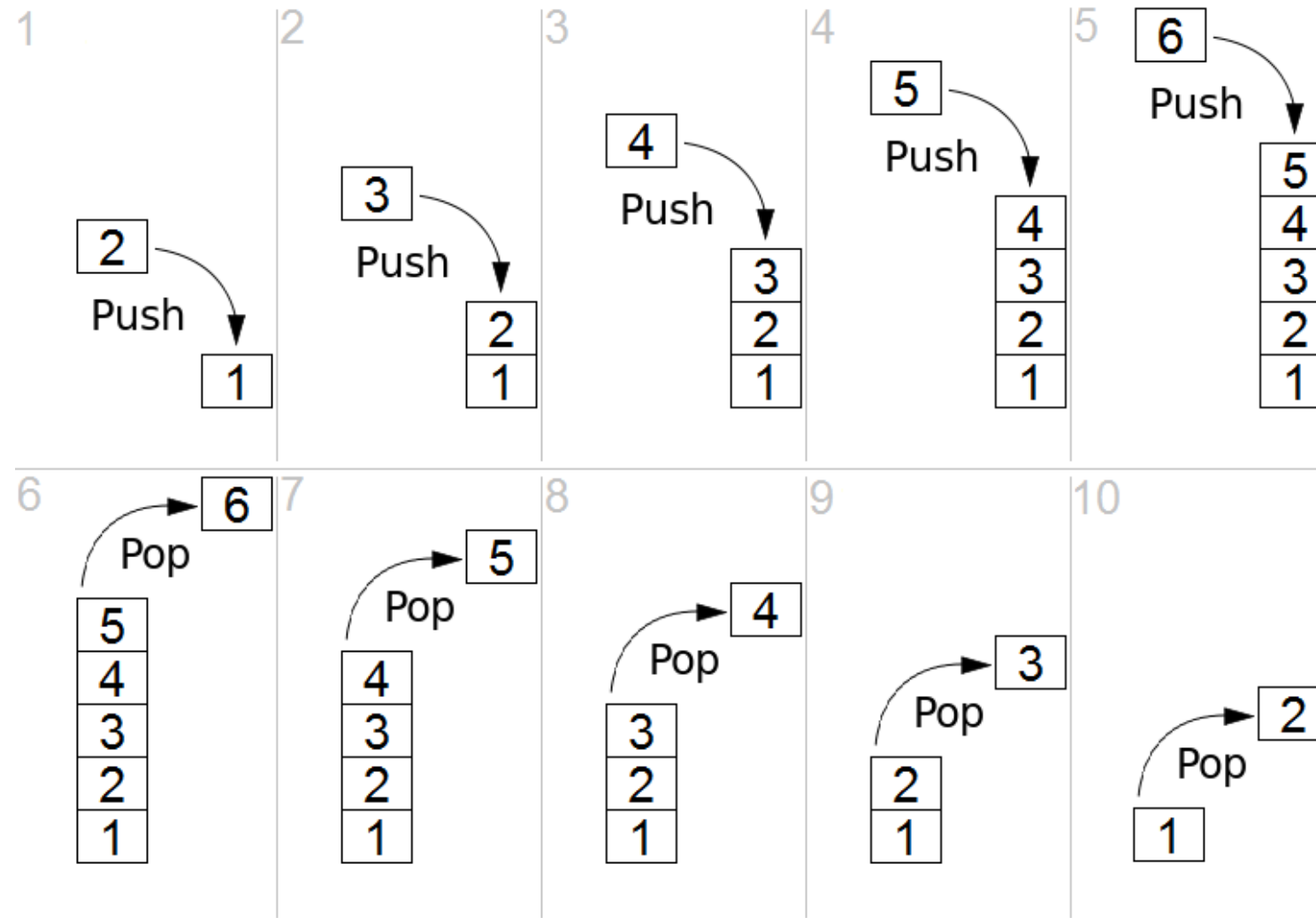
As $\mathcal{S}_\emptyset$ has no elements:

- `pop()` is not allowed
- `peek()` is not allowed.

The stack is a **LIFO** (**L**ast-**I**n, **F**irst-**O**ut) data structure, because:

- the last element that entered the stack (i.e., via `push()`)
- ...is the one that will be removed first (i.e., via `pop()`)

Effect of push() and pop() over a stack:



Simulating a Stack in Python

You can simulate the behaviour of a stack with lists.

```
S = []           # start with an empty stack
S.insert(0, 1)   # step 1 in previous slide
S.insert(0, 2)   # step 2 in previous slide
S.insert(0, 3)   # step 3 in previous slide
S.insert(0, 4)   # step 4 in previous slide
S.insert(0, 5)   # step 5 in previous slide
S.insert(0, 6)   # step 6 in previous slide
print(S)
S.pop(0)         # step 7 in previous slide
S.pop(0)         # step 8 in previous slide
S.pop(0)         # step 9 in previous slide
S.pop(0)         # step 10 in previous slide
S.pop(0)         # step 11 in previous slide
print(S)
```

```
[6, 5, 4, 3, 2, 1]
[1]
```

RECALL: `lst.insert(x, y)` inserts value `y` at position `x` in list `lst`.

NOTE: Previously, we supposed that the head of the list is the top of the stack. You can also assume that the tail of the list is the head of the stack.

```
S = []          # start with an empty stack
S.append(1)     # step 1 in slide 4
S.append(2)     # step 2 in slide 4
S.append(3)     # step 3 in slide 4
S.append(4)     # step 4 in slide 4
S.append(5)     # step 5 in slide 4
S.append(6)     # step 6 in slide 4
print(S)
S.pop()         # step 7 in slide 4
S.pop()         # step 8 in slide 4
S.pop()         # step 9 in slide 4
S.pop()         # step 10 in slide 4
S.pop()         # step 11 in slide 4
print(S)
```

```
[1, 2, 3, 4, 5, 6]
[1]
```

Queues

The **queue** is another popular abstract data type in computer science.

Queues

The **queue** is another popular abstract data type in computer science.

A queue is a sequence (collection) of elements where only the "oldest" element can be accessed.

Queues

The **queue** is another popular abstract data type in computer science.

A queue is a sequence (collection) of elements where only the "oldest" element can be accessed.

The allowed operations on a queue Q are the following:

- $\text{enqueue}(e, Q)$: adds a new element $e \in V$, for any domain V , at the beginning (back) of the queue Q
- $\text{dequeue}(Q)$: returns and removes the current element at the end (front) of the queue Q .

Queues

The **queue** is another popular abstract data type in computer science.

A queue is a sequence (collection) of elements where only the "oldest" element can be accessed.

The allowed operations on a queue Q are the following:

- $\text{enqueue}(e, Q)$: adds a new element $e \in V$, for any domain V , at the beginning (back) of the queue Q
- $\text{dequeue}(Q)$: returns and removes the current element at the end (front) of the queue Q .

Optionally, it is possible to define an additional operation:

- $\text{peek}(Q)$: reads the last element without modifying the queue.

A special domain value is $\mathcal{Q}_\emptyset$, namely the empty queue.

A special domain value is $\mathcal{Q}_\emptyset$, namely the empty queue.

As $\mathcal{Q}_\emptyset$ has no elements:

- `dequeue()` is not allowed
- `peek()` is not allowed.

A special domain value is $Q_\emptyset$, namely the empty queue.

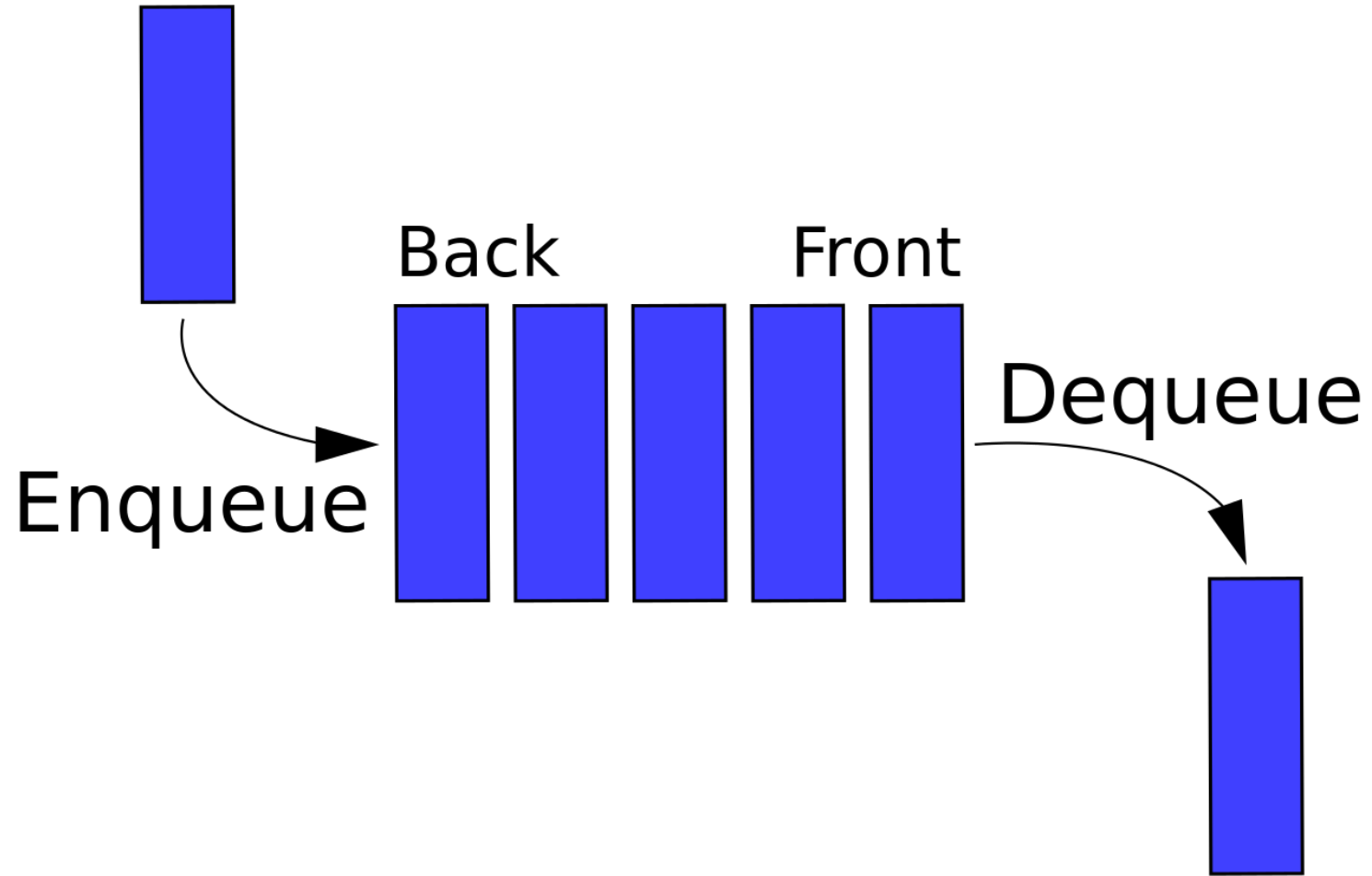
As $Q_\emptyset$ has no elements:

- `dequeue()` is not allowed
- `peek()` is not allowed.

The queue is a **FIFO** (**F**irst-**I**n, **F**irst-**O**ut) data structure, because:

- the first element that entered the queue (i.e., via `enqueue()`)
- ...is the one that will be removed first (i.e., via `dequeue()`)

Visual representation of a queue:



Simulating a Queue in Python

You can simulate the behaviour of a queue with lists.

```
Q = []           # start with an empty queue
Q.insert(0, 1)   # insert at the beginning = enqueue
Q.insert(0, 2)
Q.insert(0, 3)
Q.insert(0, 4)
Q.insert(0, 5)
Q.insert(0, 6)
print(Q)
Q.pop()          # pop at the end = dequeue
Q.pop()
Q.pop()
Q.pop()
Q.pop()
Q.pop()
print(Q)
```

```
[6, 5, 4, 3, 2, 1]
[6]
```

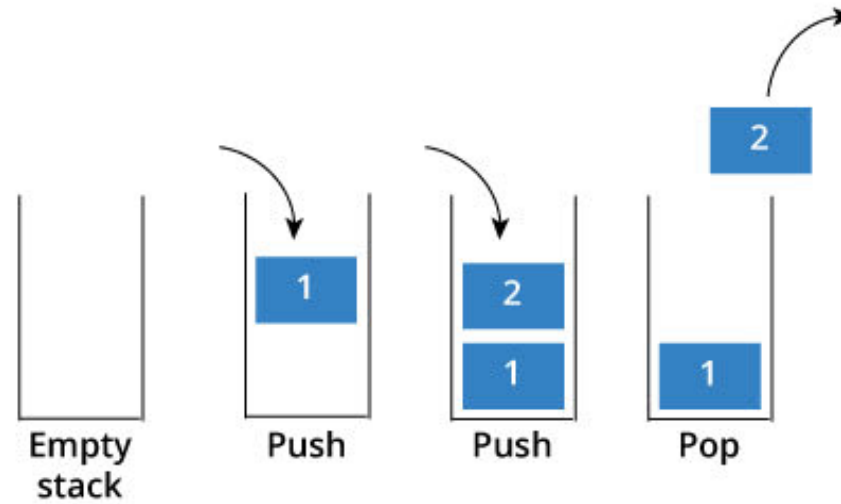
NOTE: Here we suppose that the head of the list is the beginning (back) of the queue and that the tail of the list is the end (front) of the queue.

Queue vs. Stack

Data Structure

Visual Representation

Stack



Queue



Or, more informally:

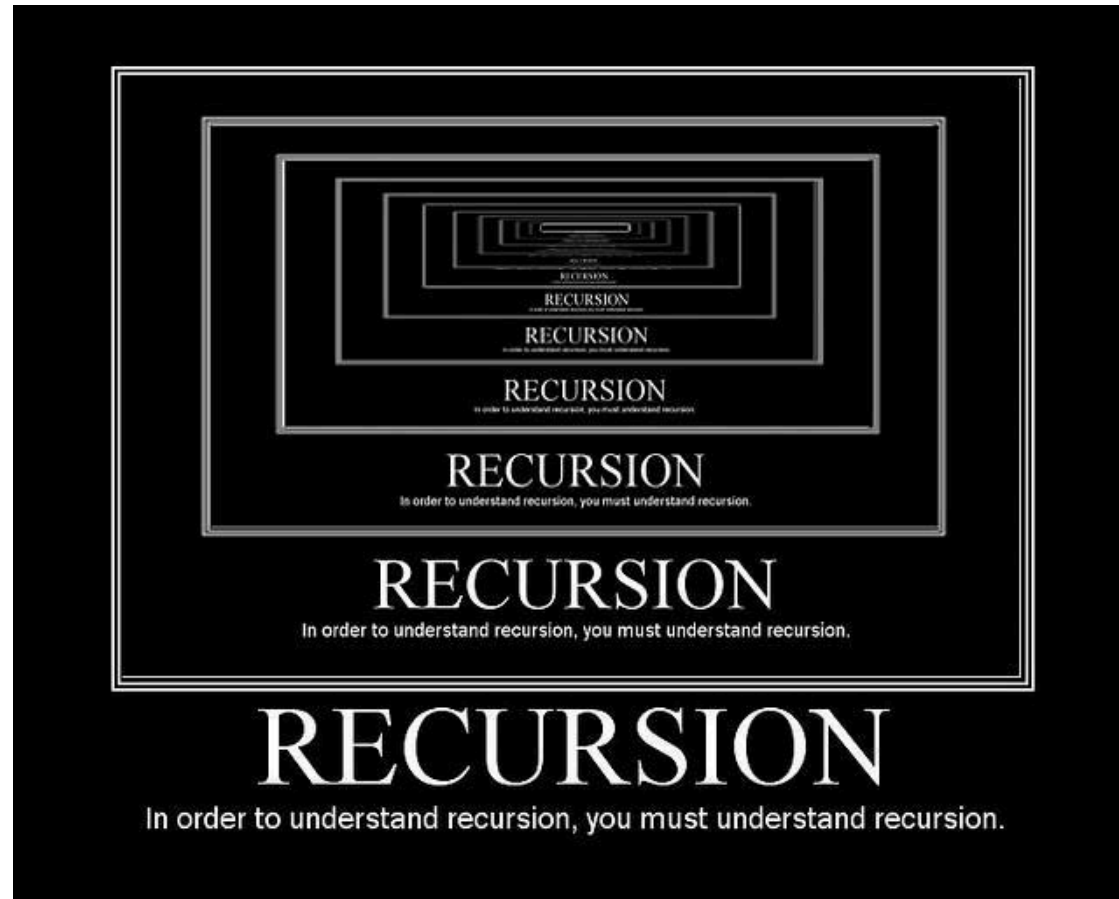
Stack



Queue



Recursion



Recursion and Problem Solving

When designing the solution of a problem

Recursion and Problem Solving

When designing the solution of a problem

...try to solve the problem assuming that you have *already solved it!*

Recursion and Problem Solving

When designing the solution of a problem

...try to solve the problem assuming that you have *already solved it!*

How is that possible?

Recursion and Problem Solving

When designing the solution of a problem

...try to solve the problem assuming that you have *already solved it!*

How is that possible?

Define the solution of a problem instance in terms of the solution of *smaller* problem instances.

Recursion and Problem Solving

When designing the solution of a problem

...try to solve the problem assuming that you have *already solved it!*

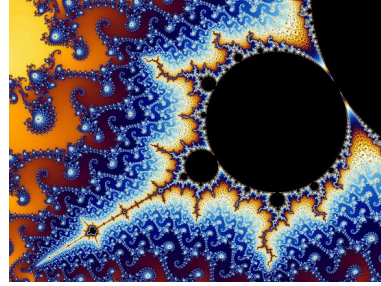
How is that possible?

Define the solution of a problem instance in terms of the solution of *smaller* problem instances.

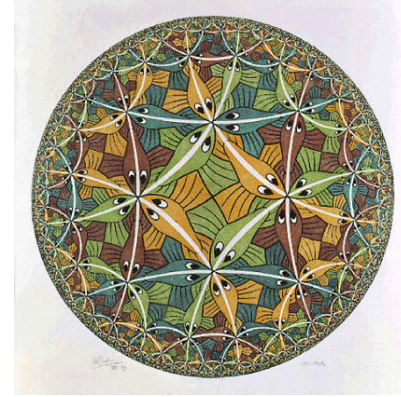
Practically speaking:

The function that computes the solution will *invoke itself* (once or more than once) from inside its body.

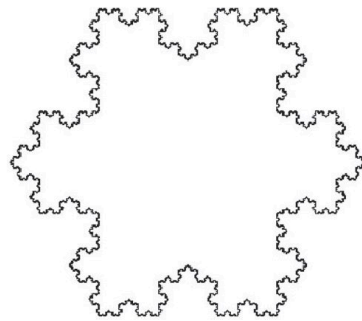
Real-world recursion



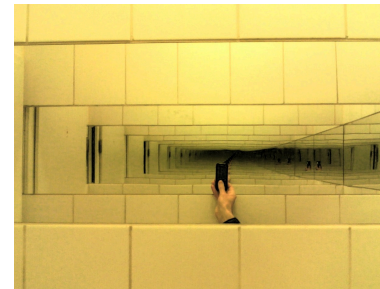
Mandelbrot Set



Escher's Circle Limit III



Koch Snowflake



Parallel Mirrors

A more practical example? Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

A more practical example? Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

Back-of-the-envelope calculation: $5!$

$$\begin{array}{c} 5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot \underbrace{1}_{1!} \\ \underbrace{\hspace{1.5cm}}_{2 \cdot 1!} \\ \underbrace{\hspace{1.5cm}}_{3 \cdot 2!} \\ \underbrace{\hspace{1.5cm}}_{4 \cdot 3!} \\ \underbrace{\hspace{1.5cm}}_{5 \cdot 4!} \end{array}$$

A more practical example? Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

Back-of-the-envelope calculation: $5!$

$$\begin{array}{c} 5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot \underbrace{1}_{1!} \\ \underbrace{\hspace{1.5cm}}_{2 \cdot 1!} \\ \underbrace{\hspace{1.5cm}}_{3 \cdot 2!} \\ \underbrace{\hspace{1.5cm}}_{4 \cdot 3!} \\ \underbrace{\hspace{1.5cm}}_{5 \cdot 4!} \end{array}$$

We can spot the following pattern:

$$n! = n \cdot (n - 1)!$$

with base case:

$$0! = 1$$

Recursion and Problem Solving (cont'd)

In order to use a recursive approach:

1. you must be able to **decompose** the problem into problems of the same type
2. you must find a **relation** that connects a given problem to the solution of similar subproblems
3. you need to know the solution of a special case of the problem (the **base case**, where the recursion terminates).

Recursion and Problem Solving (cont'd)

In order to use a recursive approach:

1. you must be able to **decompose** the problem into problems of the same type
2. you must find a **relation** that connects a given problem to the solution of similar subproblems
3. you need to know the solution of a special case of the problem (the **base case**, where the recursion terminates).

Take the factorial example:

1. multiplication between two numbers
2. multiplication between n and the already-known $(n - 1)!$
3. the base case is $0! = 1$

Approach for Problem Decomposition

When solving a task with a recursive approach:

Approach for Problem Decomposition

When solving a task with a recursive approach:

1. **Identify the base case(s):** Determine the simplest version of the problem that can be solved directly. For instance: we know that the factorial of 0 is 1, i.e. $0! = 1$

Approach for Problem Decomposition

When solving a task with a recursive approach:

1. **Identify the base case(s):** Determine the simplest version of the problem that can be solved directly. For instance: we know that the factorial of 0 is 1, i.e. $0! = 1$
2. **Define the recursive case:** Break down the problem into a smaller version of the same problem. For instance, the factorial of a number n can be computed by computing *factorial*($n - 1$).

Approach for Problem Decomposition

When solving a task with a recursive approach:

1. **Identify the base case(s):** Determine the simplest version of the problem that can be solved directly. For instance: we know that the factorial of 0 is 1, i.e. $0! = 1$
2. **Define the recursive case:** Break down the problem into a smaller version of the same problem. For instance, the factorial of a number n can be computed by computing *factorial*($n - 1$).
3. **Ensure progress toward the base case:** Make sure each recursive call moves closer to the base case. For instance, the recursive call in the factorial function moves closer to the base case by decreasing the value of n by 1 in each recursive call.

Approach for Problem Decomposition

When solving a task with a recursive approach:

1. **Identify the base case(s):** Determine the simplest version of the problem that can be solved directly. For instance: we know that the factorial of 0 is 1, i.e. $0! = 1$
2. **Define the recursive case:** Break down the problem into a smaller version of the same problem. For instance, the factorial of a number n can be computed by computing *factorial*($n - 1$).
3. **Ensure progress toward the base case:** Make sure each recursive call moves closer to the base case. For instance, the recursive call in the factorial function moves closer to the base case by decreasing the value of n by 1 in each recursive call.
4. **Combine the results:** Determine how to combine the result of the recursive call with any additional processing needed. For instance, in the factorial function, we need to multiply the result of the recursive call by the current value of n .

Approach for Problem Decomposition

When solving a task with a recursive approach:

1. **Identify the base case(s):** Determine the simplest version of the problem that can be solved directly. For instance: we know that the factorial of 0 is 1, i.e. $0! = 1$
2. **Define the recursive case:** Break down the problem into a smaller version of the same problem. For instance, the factorial of a number n can be computed by computing *factorial*($n - 1$).
3. **Ensure progress toward the base case:** Make sure each recursive call moves closer to the base case. For instance, the recursive call in the factorial function moves closer to the base case by decreasing the value of n by 1 in each recursive call.
4. **Combine the results:** Determine how to combine the result of the recursive call with any additional processing needed. For instance, in the factorial function, we need to multiply the result of the recursive call by the current value of n .
5. **Verify correctness:** Test with simple examples to ensure your recursive solution works properly.

Issues with Recursion

There are two major issues with recursion:

Issues with Recursion

There are two major issues with recursion:

1. **Infinite recursion**

- missing base case?
- function called with wrong arguments? For example, what happens if we do $(-1)!$?
...more on this later

Issues with Recursion

There are two major issues with recursion:

1. **Infinite recursion**

- missing base case?
- function called with wrong arguments? For example, what happens if we do $(-1)!$?
...more on this later

2. **Un-responsiveness**

- your code can easily become very slow, also with *easy* input parameters
- ... more on memory management later

Issues with Recursion

There are two major issues with recursion:

1. **Infinite recursion**

- missing base case?
- function called with wrong arguments? For example, what happens if we do $(-1)!$?
...more on this later

2. **Un-responsiveness**

- your code can easily become very slow, also with *easy* input parameters
- ... more on memory management later

NOTE: Python is quite good at catching infinite recursions: it has a dedicated error: **RecursionError: maximum recursion depth exceeded in comparison**

Keeping track of recursive calls

To trace recursive function calls, we need to use a **stack** data structure:

- Each time a recursive function is called, a new frame is pushed onto the call stack, containing the function's parameters and local variables.
- When a function returns, its frame is popped off the stack.

This stack-based mechanism allows the program to *remember* where to return after each recursive call completes.

Example: countdown

Write a recursive function `countdown(n)` that takes a positive integer `n` and prints the countdown. As the countdown reaches 0, prints `Blastoff!` and quits.

1. **Identify the base case(s):** when $n = 0$, print Blastoff!

1. **Identify the base case(s):** when $n = 0$, print Blastoff!
2. **Identify the recursive case(s):** when $n > 0$, print n and call the function with $n - 1$

1. **Identify the base case(s):** when $n = 0$, print Blastoff!
2. **Identify the recursive case(s):** when $n > 0$, print n and call the function with $n - 1$
3. **Ensure progress toward the base case:** the recursive call is with $n - 1$, which is smaller than n and will eventually reach 0, our base case.

```
def countdown(n):  
    if n==0:  
        print("Blastoff!")  
    else:  
        print(n)  
        countdown(n-1)  
  
countdown(3)
```

```
3  
2  
1  
Blastoff!
```

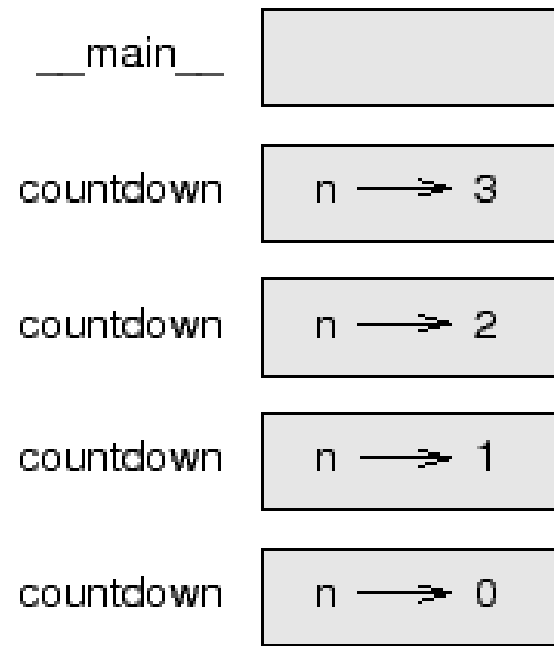
What's going on here?

- (1) The execution of countdown begins with $n=3$, and since n is greater than 0, it outputs the value 3, and then calls itself...
 - (2) The execution of countdown begins with $n=2$, and since n is greater than 0, it outputs the value 2, and then calls itself...
 - (3) The execution of countdown begins with $n=1$, and since n is greater than 0, it outputs the value 1, and then calls itself...
 - (4) The execution of countdown begins with $n=0$, and since n is not greater than 0, it outputs Blastoff! and then returns.
 - (5) The countdown that got $n=1$ returns.
 - (6) The countdown that got $n=2$ returns.
 - (7) The countdown that got $n=3$ returns. And we are back into the main script.

RECALL: Every time a function gets called, Python creates a new function frame, which contains the function's local variables and parameters.

For a recursive function, there might be more than one frame at the same time. Python organises those frames as a **stack**.

For instance, the stack diagram for `countdown(3)` is:



How does the stack in the figure works?

1. (as usual) at the top of the stack is the frame for the `main`*
2. we call `countdown(3)`, so there is a new frame in the stack
3. `countdown(3)` calls `countdown(2)`, so there is a new frame in the stack
4. `countdown(2)` calls `countdown(1)`, so there is a new frame in the stack
5. `countdown(1)` calls `countdown(0)`, so there is a new frame in the stack
6. `countdown(0)` does not make any further calls, so no more frames

* In our case, it is empty because we did not create any variables in `main` or pass any arguments to it.

How does the stack in the figure works?

1. (as usual) at the top of the stack is the frame for the `main`*
2. we call `countdown(3)`, so there is a new frame in the stack
3. `countdown(3)` calls `countdown(2)`, so there is a new frame in the stack
4. `countdown(2)` calls `countdown(1)`, so there is a new frame in the stack
5. `countdown(1)` calls `countdown(0)`, so there is a new frame in the stack
6. `countdown(0)` does not make any further calls, so no more frames

* In our case, it is empty because we did not create any variables in `main` or pass any arguments to it.

We did a bunch of pushes, now we must do some pops:

Exercises

factorial

Write a recursive function `factorial(n)` that takes as input a positive integer `n` and returns $n!$

factorial

Write a recursive function `factorial(n)` that takes as input a positive integer `n` and returns $n!$

Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

factorial

Write a recursive function `factorial(n)` that takes as input a positive integer `n` and returns $n!$

Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

Hence:

factorial

Write a recursive function `factorial(n)` that takes as input a positive integer n and returns $n!$

Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

Hence:

- **Base case:**
 - When $n = 0$, then 1
 - This comes from: $0! = 1$

factorial

Write a recursive function `factorial(n)` that takes as input a positive integer n and returns $n!$

Computing the factorial of a number n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 1$$

Hence:

- **Base case:**
 - When $n = 0$, then 1
 - This comes from: $0! = 1$
- **Recursive case:**
 - otherwise we know that the factorial of n is n times the factorial of $n - 1$
 - This comes from $n! = n \cdot (n - 1)!$

```
def factorial(n):  
    if n==0:  
        return 1  
    else:  
        return n * factorial(n-1)  
  
output = factorial(5)  
print(output)
```

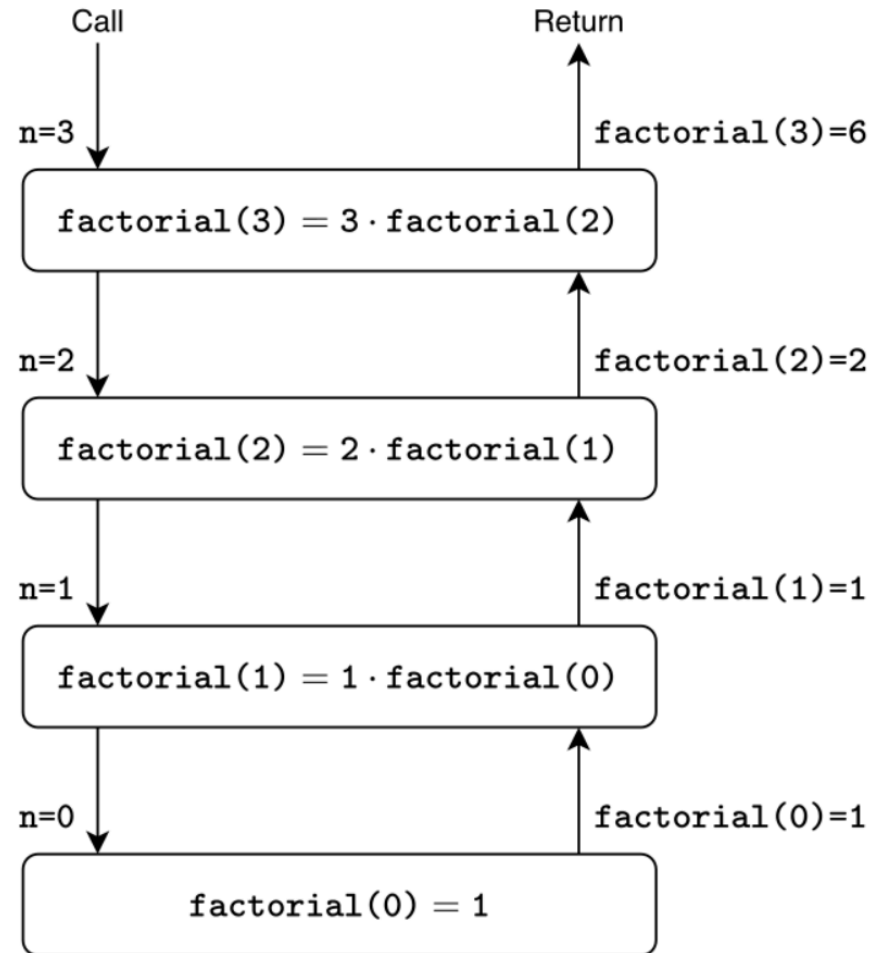
120

```
def factorial(n):  
    if n==0:  
        return 1  
    else:  
        return n * factorial(n-1)  
  
output = factorial(5)  
print(output)
```

120

WARNING: What happens when calling factorial(-1)?

Stack diagram for `factorial(3)`:



palindrome

Write a recursive function `palindrome(s)` that takes as input a string `s` and returns `True` if `s` is palindrome and `False` otherwise.

palindrome

Write a recursive function `palindrome(s)` that takes as input a string `s` and returns `True` if `s` is palindrome and `False` otherwise.

Decomposition strategy: let `s` be "racecar"

- racecar -> same letter ✓
- aceca -> same letter ✓
- cec -> same letter ✓
- e -> trivial ✓

palindrome

Write a recursive function `palindrome(s)` that takes as input a string `s` and returns `True` if `s` is palindrome and `False` otherwise.

Decomposition strategy: let `s` be "racecar"

- racecar -> same letter ✓
- aceca -> same letter ✓
- cec -> same letter ✓
- e -> trivial ✓

and now let `s` be "rediviuer"

- rediviuer -> same letter ✓
- ediviue -> same letter ✓
- diviu -> different letter ✗

Hence, "racecar" is palindrome, but "rediviuer" it's not.

From what we have seen so far:

From what we have seen so far:

- **Base case:**
 - when the string is empty or 1 character long, it is a palindrome

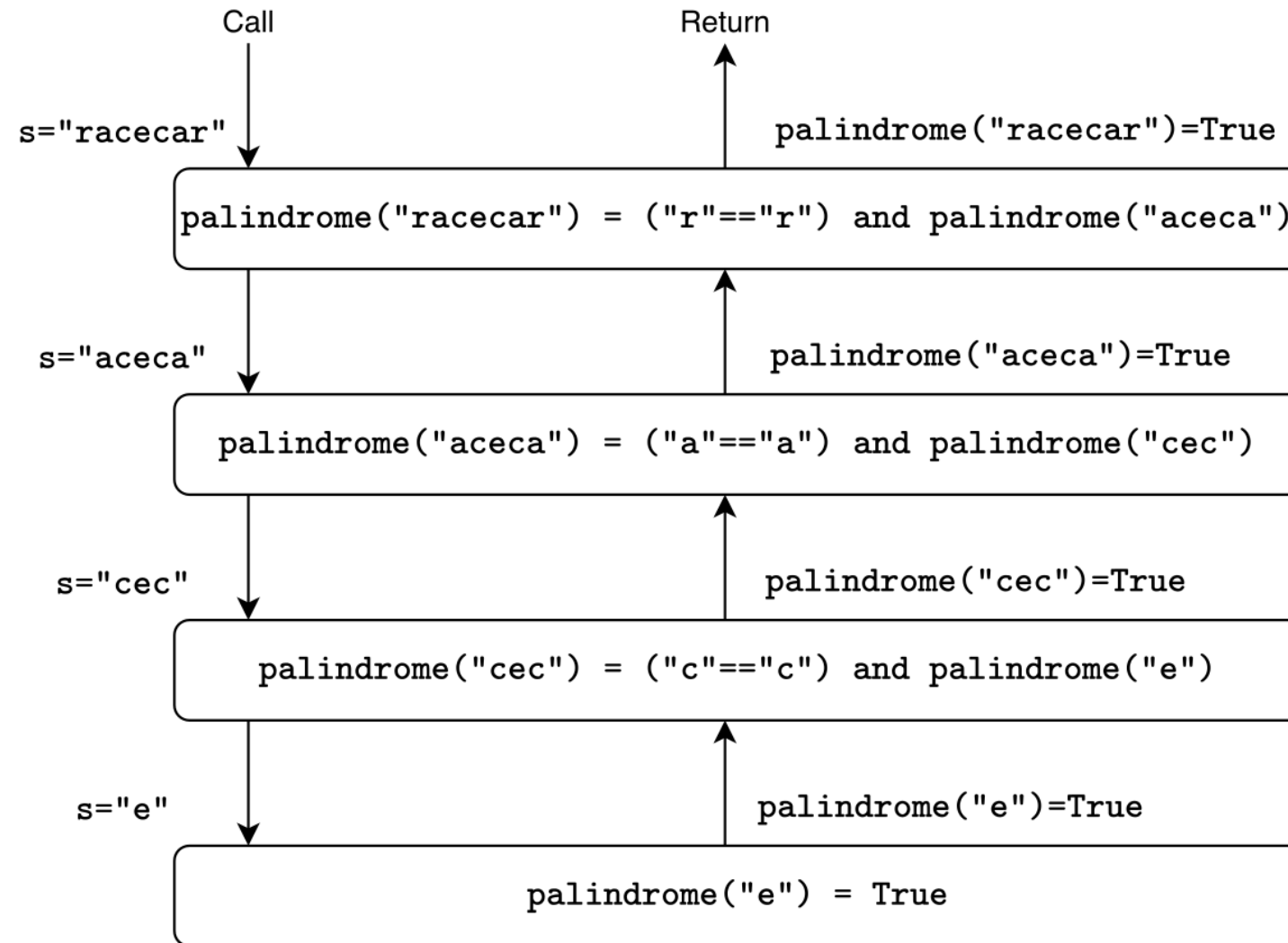
From what we have seen so far:

- **Base case:**
 - when the string is empty or 1 character long, it is a palindrome
- **Recursive case:**
 - otherwise, check if the first and last characters are the same
 - if they are, remove the first and last characters and recursively check the remaining string
 - if they are not, it is not a palindrome

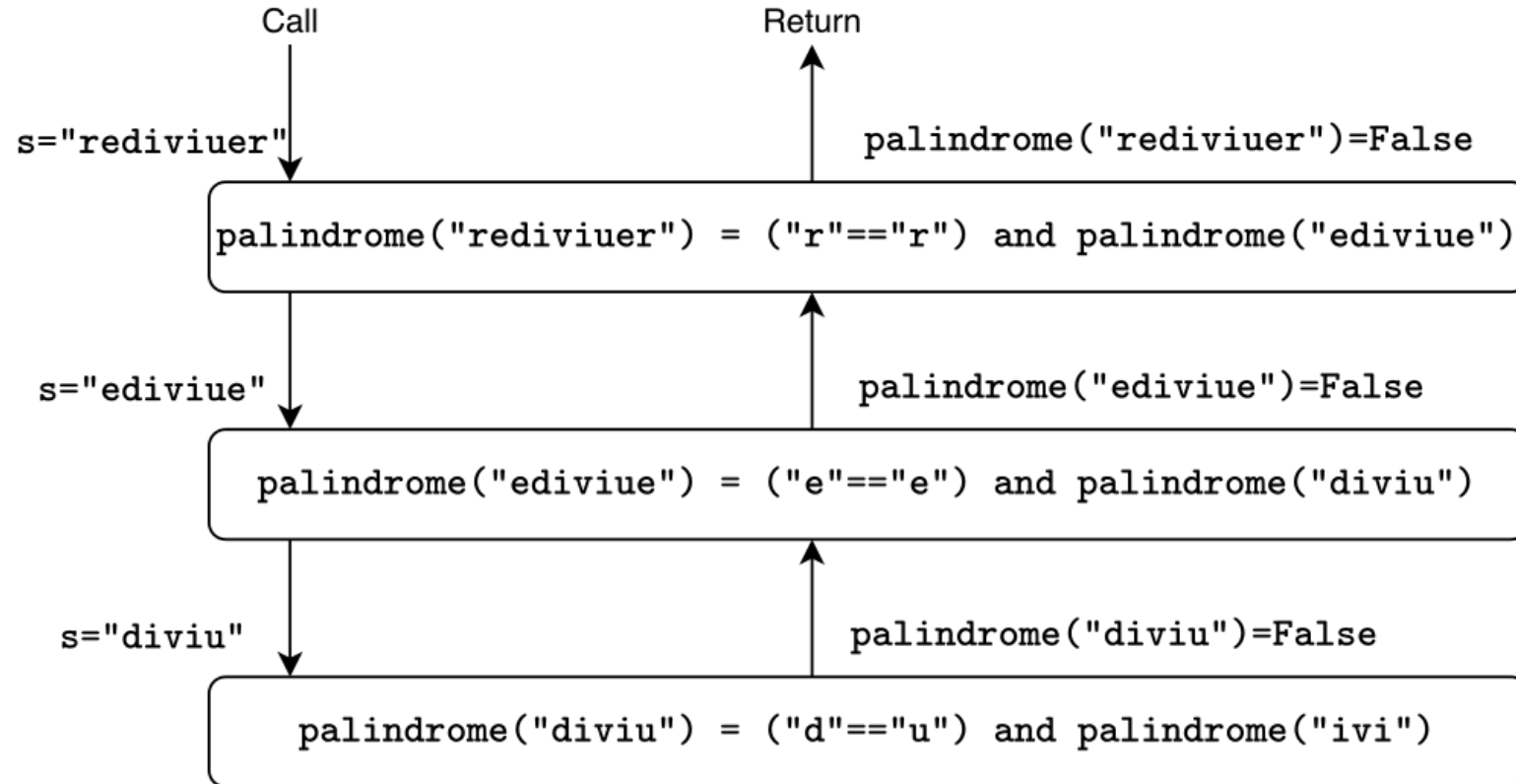
```
def palindrome(s):  
    if len(s) <= 1:  
        return True  
    else:  
        return (s[0] == s[-1]) and palindrome(s[1:-1])  
  
output1 = palindrome("racecar")  
print(output1)  
output2 = palindrome("rediviuer")  
print(output2)
```

True
False

Stack diagram for `palindrome("racecar")`:



Stack diagram for `palindrome("rediviuer")`:



fibonacci

Write a recursive function `fibonacci(n)` that takes as input a positive integer `n` and returns the n -th Fibonacci number.

fibonacci

Write a recursive function `fibonacci(n)` that takes as input a positive integer n and returns the n -th Fibonacci number.

RECALL: Each number in the Fibonacci sequence is given by the sum of the two preceding ones:

$$F(n) = F(n - 1) + F(n - 2)$$

where $F(0) = 0$ and $F(1) = 1$.

Hence: 0 1 1 2 3 5 8 13 ...

fibonacci

Write a recursive function `fibonacci(n)` that takes as input a positive integer n and returns the n -th Fibonacci number.

RECALL: Each number in the Fibonacci sequence is given by the sum of the two preceding ones:

$$F(n) = F(n - 1) + F(n - 2)$$

where $F(0) = 0$ and $F(1) = 1$.

Hence: 0 1 1 2 3 5 8 13 ...

Decomposition strategy: `fibonacci(n)` with $n=5$ *:

$$\begin{array}{c} \underbrace{\quad\quad\quad}_5 \\ \underbrace{\quad}_3 \quad + \quad \underbrace{\quad}_2 \\ \underbrace{\quad}_2 \quad + \quad 1 \quad \quad 1 \quad + \quad \underbrace{\quad}_2 \\ \underbrace{\quad}_2 \quad + \quad 1 \quad \quad 1 \quad + \quad 1 \\ 1 \quad + \quad 1 \end{array}$$

* By coincidence, the 5th Fibonacci number is 5.

From what we have seen so far:

From what we have seen so far:

- **Base case:**

- When $n = 0$, then $f(0) = 0$
- When $n = 1$, then $f(1) = 1$

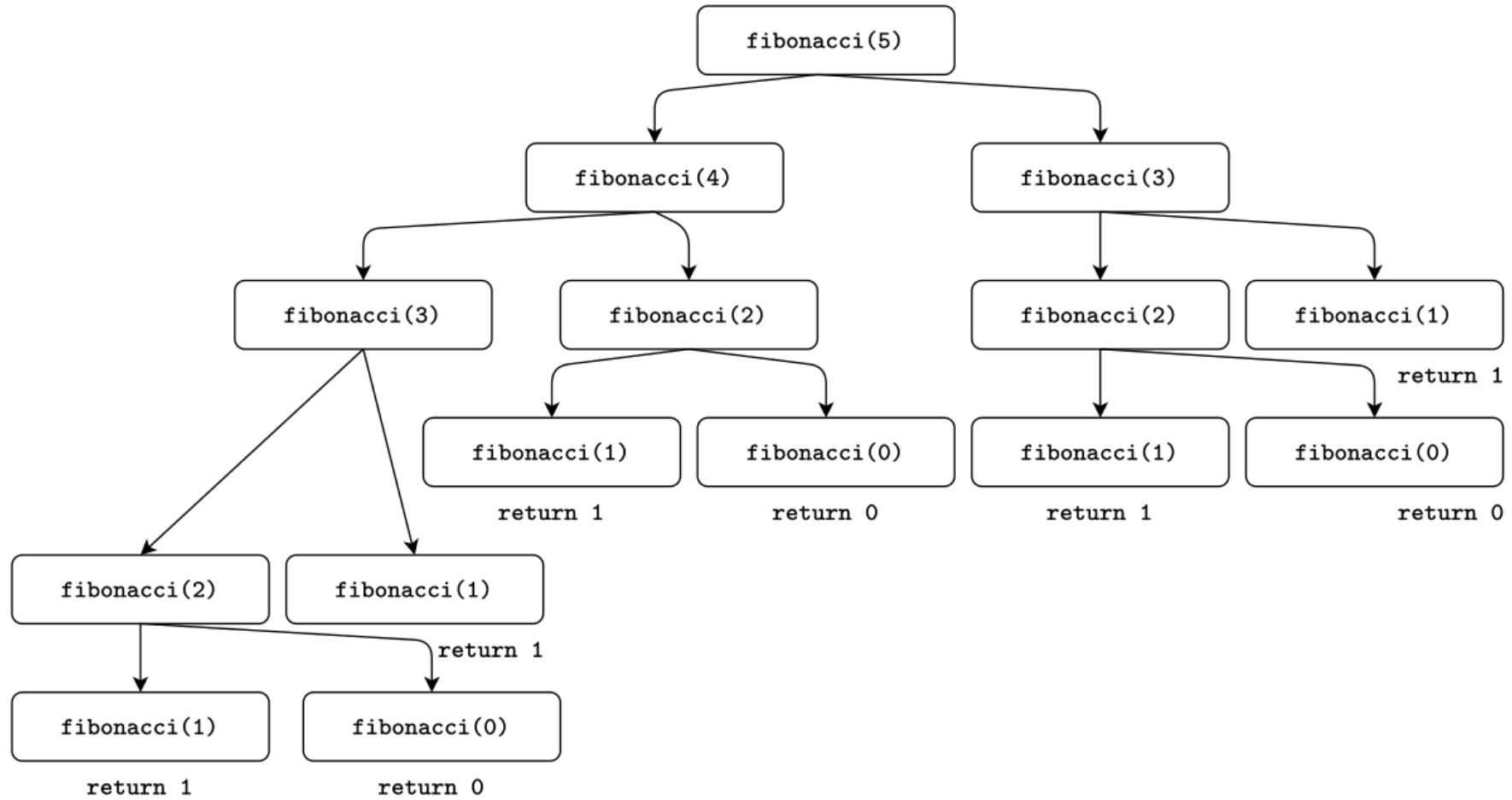
- **Recursive case:**

- Otherwise the fibonacci number is the sum of the previous two numbers, i.e.,
$$f(n) = f(n - 1) + f(n - 2)$$

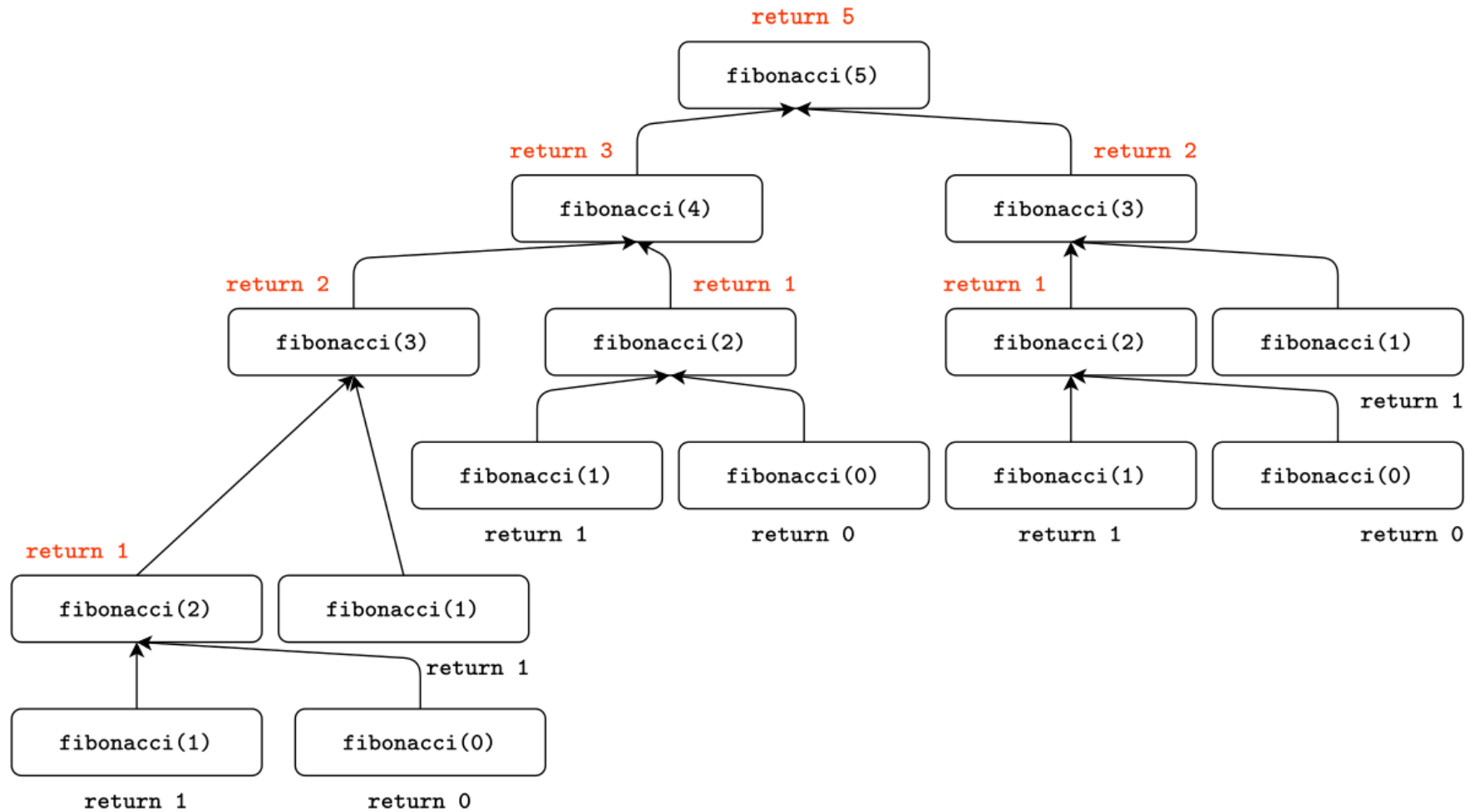
```
def fibonacci(n):  
    if n in [0, 1]:  
        return n  
    else:  
        return (fibonacci(n-1) + fibonacci(n-2))  
  
output = fibonacci(5)  
print(output)
```

5

Call Stack diagram for fibonacci(5):



Return Stack diagram for fibonacci(5):



Exercise session

Find the exercises for the exam session: [here](#)

